

PROBLEM STATEMENT

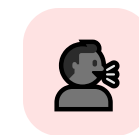
The Mobility Booking Friction in Jordan

Booking a ride still means typing — and global maps don't speak Amman.



Typing on the go is slow and unsafe

Entering addresses while walking, carrying bags or holding a child means mistakes, delays and eyes off the street.



Global maps don't understand Jordanian Arabic

Local slang and landmark names — **دوار الواحة، الكيلو، السابع** — are what people say, not what map search expects.



Ambiguous requests

Riders ask for «**الجامعة**» or «**المستشفى**» without saying which one — a guess sends the car to the wrong place.



Wrong gate, failed pickup

Mecca Mall, City Mall and the University of Jordan have several gates; a pin on the building centre means missed pickups and lost fares.

SOLUTION & SCOPE

PetraVoice — Conversational AI Mobility Assistant

- ✓ **Hands-free voice booking** built for Petra Ride riders in Jordan — tap once and speak.
- ✓ **Authentic Jordanian dialect** (Ammani colloquial) and English: «يعطيك العافية، بدي سيارة من مكة مول بوابة 2 لدوار صويلح»
- ✓ **Automatic gate pinpointing** — «بوابة 2» lands on the exact road-side gate, not the roof.
- ✓ **Smart disambiguation** — a generic «المستشفى» or «الجامعة» gets the top 5 Jordanian options, spoken and on screen.
- ✓ **Ride tiers & live pricing** — Economy, Comfort and Family XL priced per trip; a tier said out loud is pre-selected.
- ✓ **Safety by design** — the app reads the ride back, but books only after the rider taps confirm.



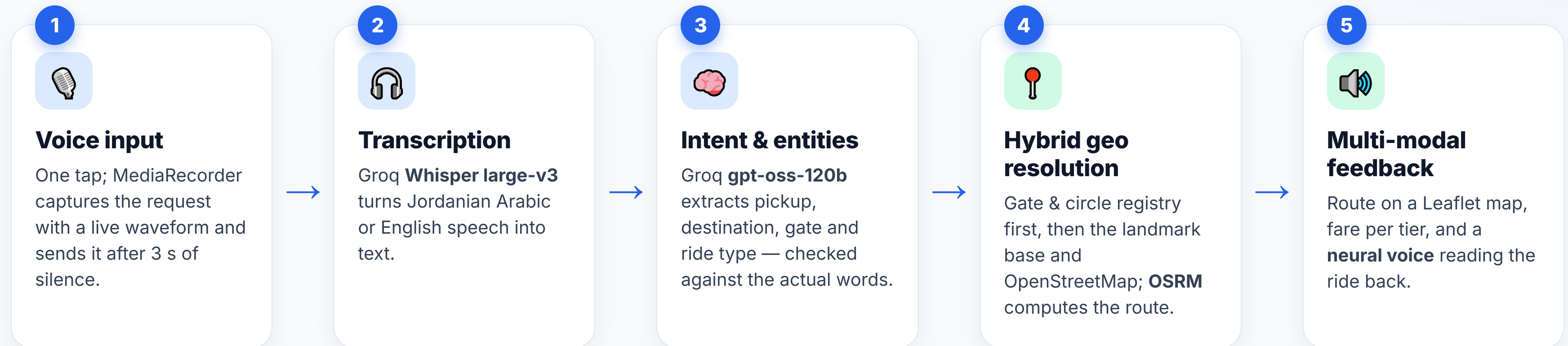
Mecca Mall (Gate 2) → Sweileh Circle, priced



«أي جامعة؟» — five options

END-TO-END USER WORKFLOW

How the User Experience Flows

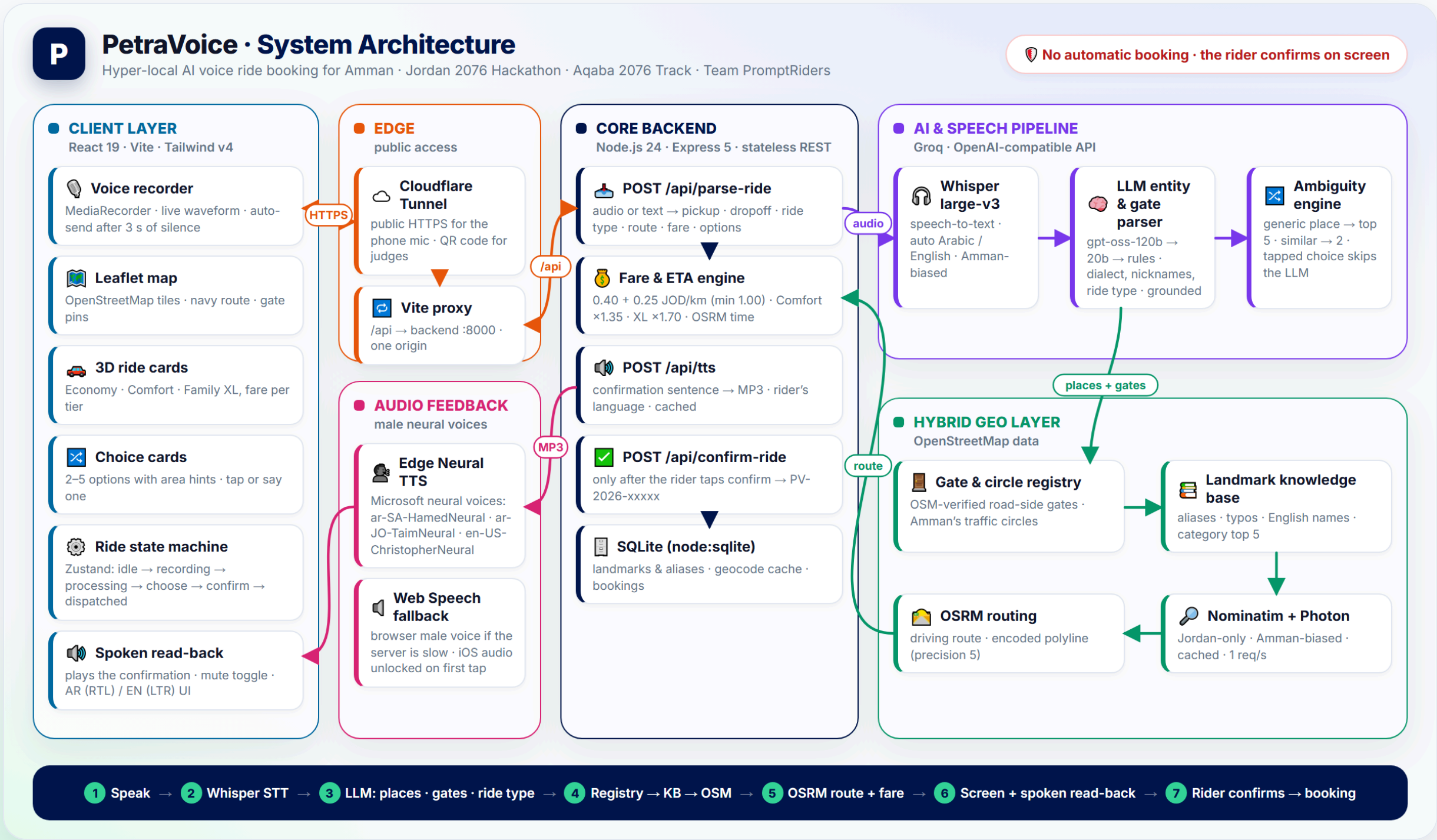


Vague place? «وصلني ع المستشفى» → spoken «أبي مستشفى؟» + 5 cards → one tap → route in **0.4 s** (no extra AI call)

Rider confirms → booking

SYSTEM ARCHITECTURE

Technical Architecture & Intelligence Stack



Frontend client
React 19 · Vite · Tailwind CSS · Leaflet · MediaRecorder API

AI cloud engine
Groq: Whisper large-v3 + gpt-oss-120b, with a 20b model and a rule parser as fallbacks

Backend engine
Node.js 24 · Express 5 · fare & ETA engine · SQLite · Cloudflare Tunnel

Geo intelligence
Curated gate & circle registry + OpenStreetMap (Jordan-only) + OSRM routing

Audio feedback
Microsoft Edge neural TTS: ar-JO-TaimNeural / ar-SA-HamedNeural · browser fallback

LIVE DEMO

Try PetraVoice Live



yusefalsalman.github.io/PetraVoice

Scan with your phone camera · allow the microphone

Code: github.com/yusefalsalman/PetraVoice

Say one of these after tapping the mic

«بدي سيارة من مكة مول بوابة 2 لدوار صويلح»

exact gate pin + circle

«وصلني على الجامعة الأردنية البوابة الشمالية»

gate from your current location

«بدي أروح على المستشفى»

5-choice disambiguation

"Take me from City Mall to the 7th Circle"

English works too

One tap → the app listens, maps, prices and reads the ride back. Nothing is booked until you tap **Confirm**.

TESTING, EDGE CASES & ROADMAP

Validation, Current Limitations & Next Steps

✓ What we tested

- Automated end-to-end suite: **9 / 9 passing**, every pin checked to within 30 m
- Gate pins: Mecca Mall (Gate 2), UJ North Gate; circles incl. Sweileh and الكيلو
- Dialect, fillers and nicknames; English requests; ride type from speech
- Generic places: 5 hospitals / 5 malls; tapped choice answered without the LLM
- Real speech through the mic pipeline — Whisper transcripts word-for-word

⚠ Current limitations

- Needs an internet connection (Groq, OpenStreetMap, OSRM, voice)
- Free-tier AI rate limits; a rule-based parser keeps text requests working
- Curated gates cover key venues; the 6th Circle and three schools await mapping
- Mecca Mall gate numbering to verify on site

🚀 Roadmap

- Native SDK integration into the Petra Ride passenger app
- Crowd-sourced gate & entrance registry for more venues
- Offline-cached intents for frequent trips
- Driver-side voice navigation to the exact gate
- Aqaba and other cities